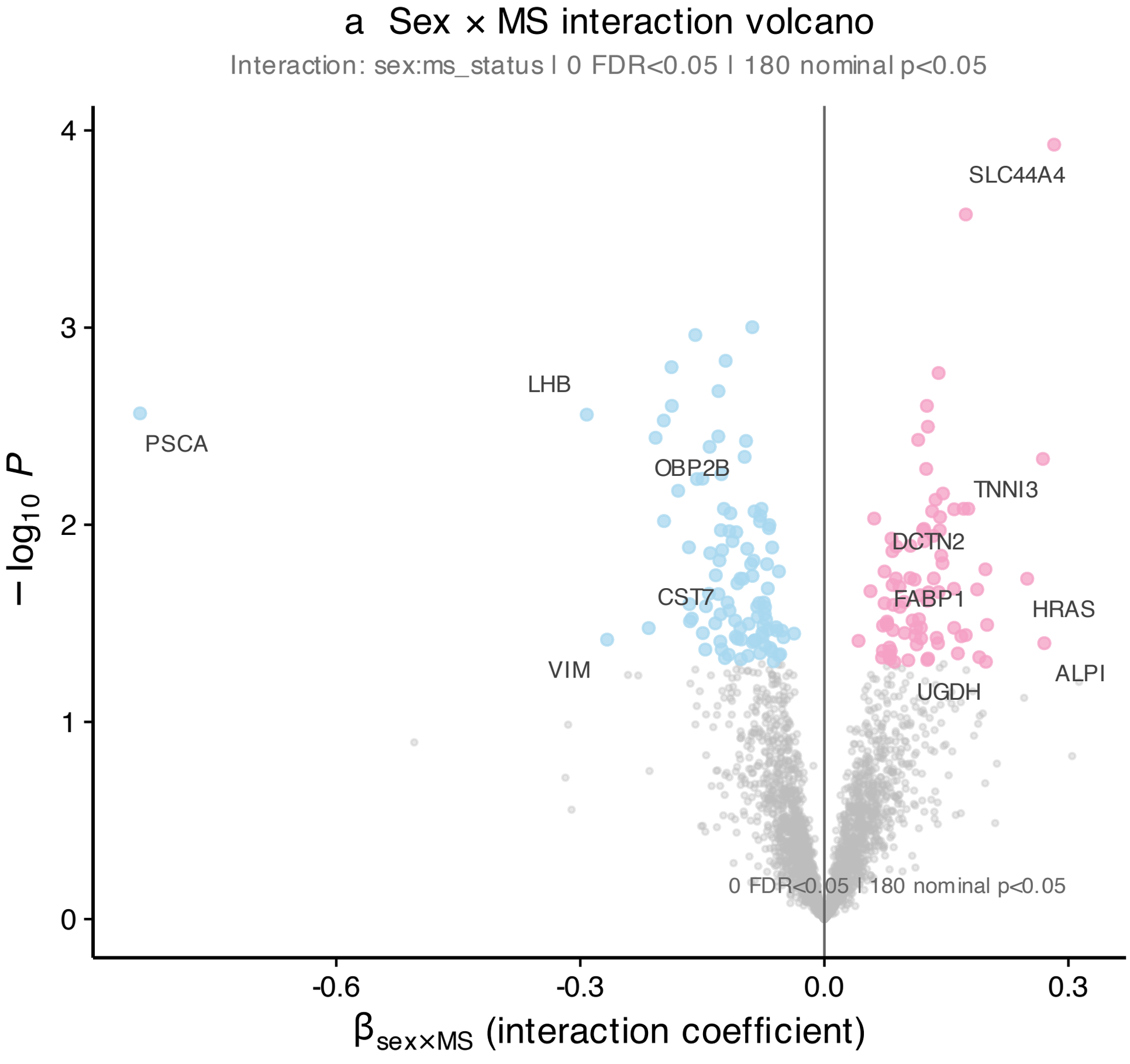
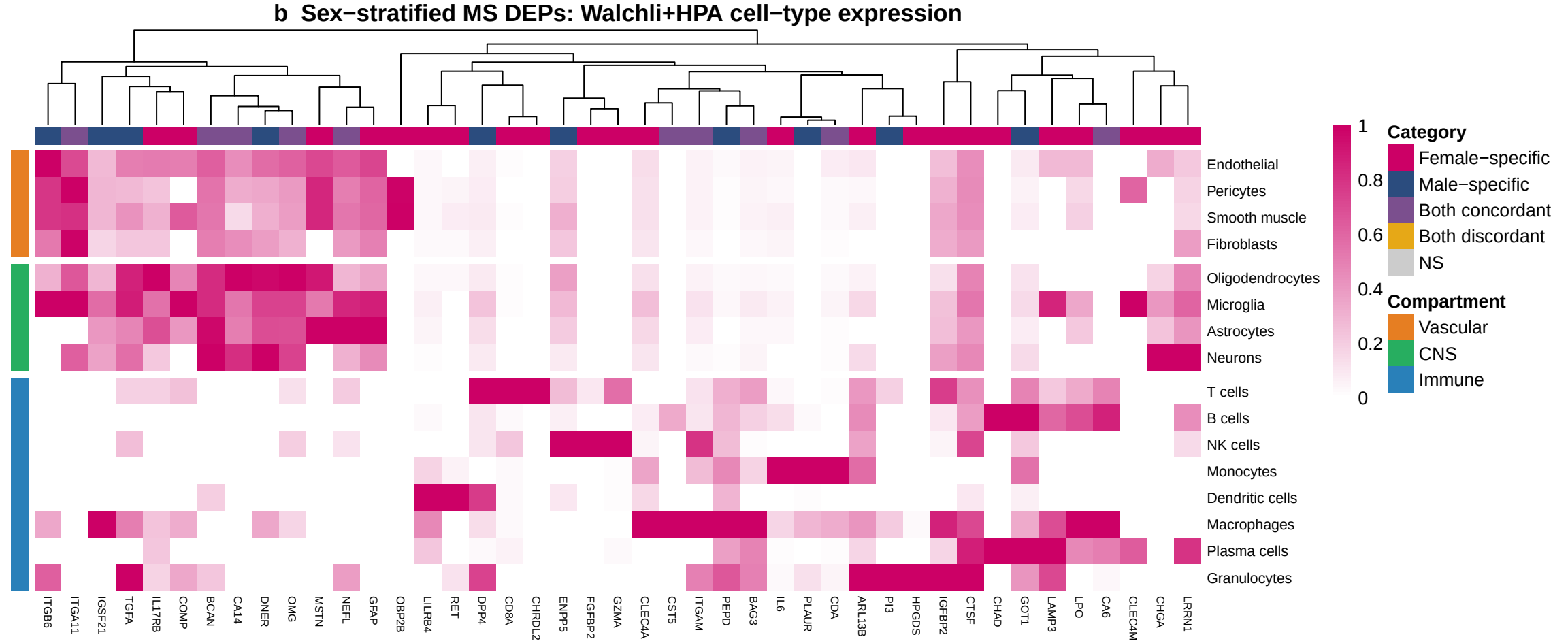
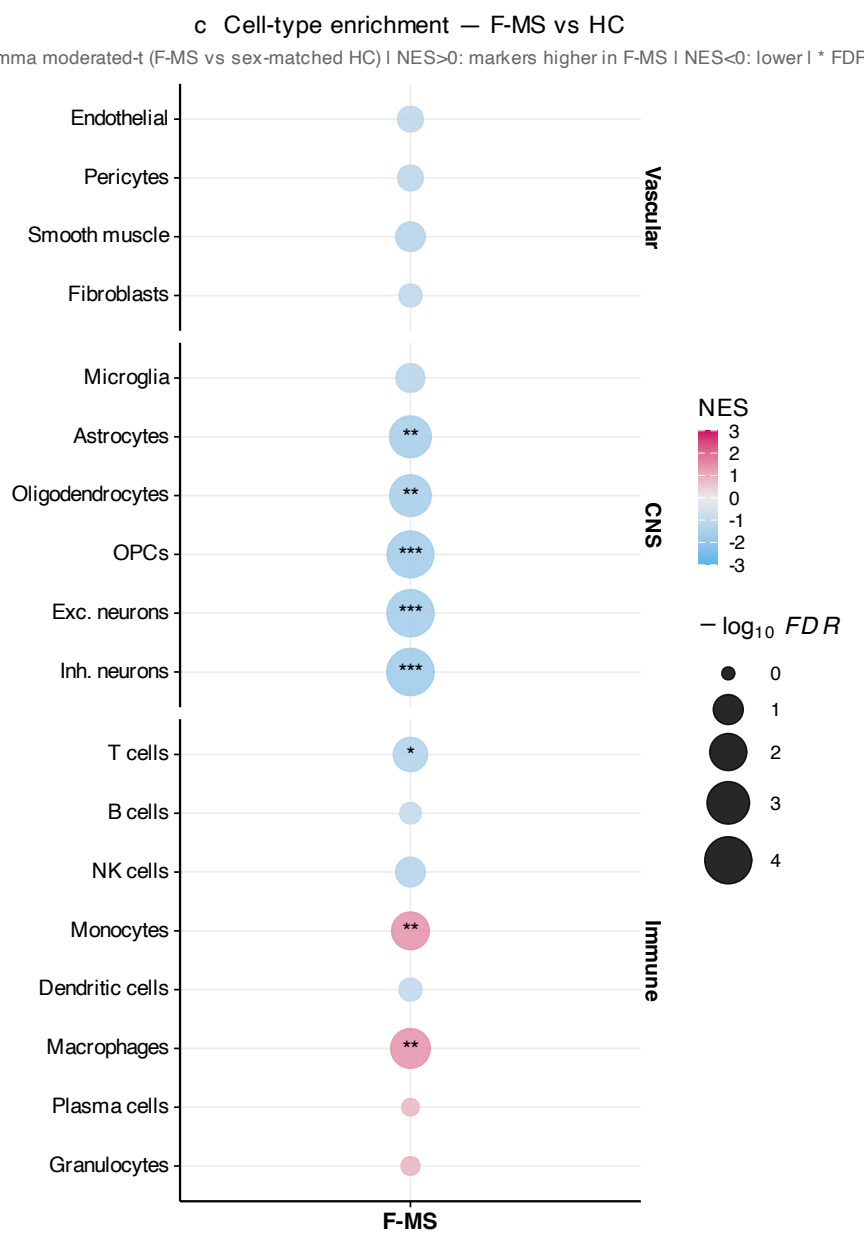
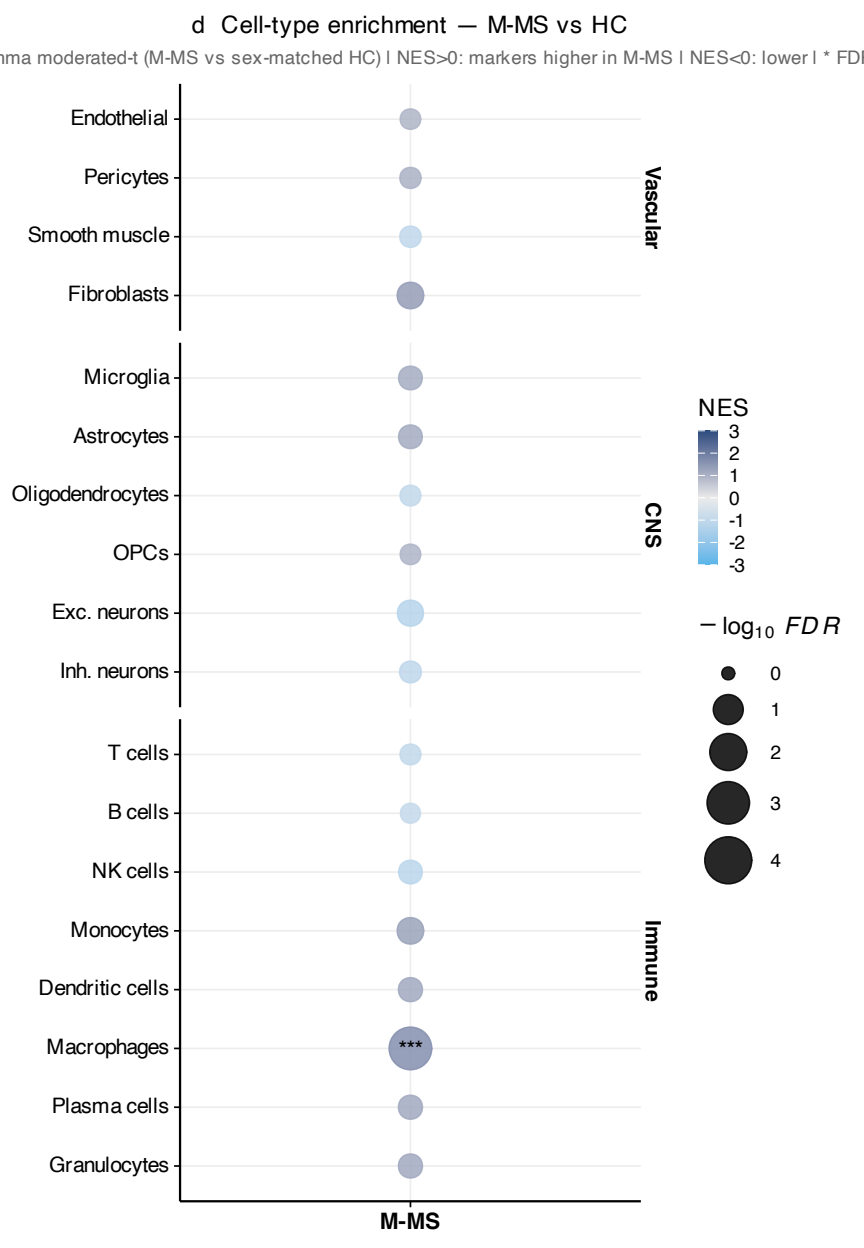
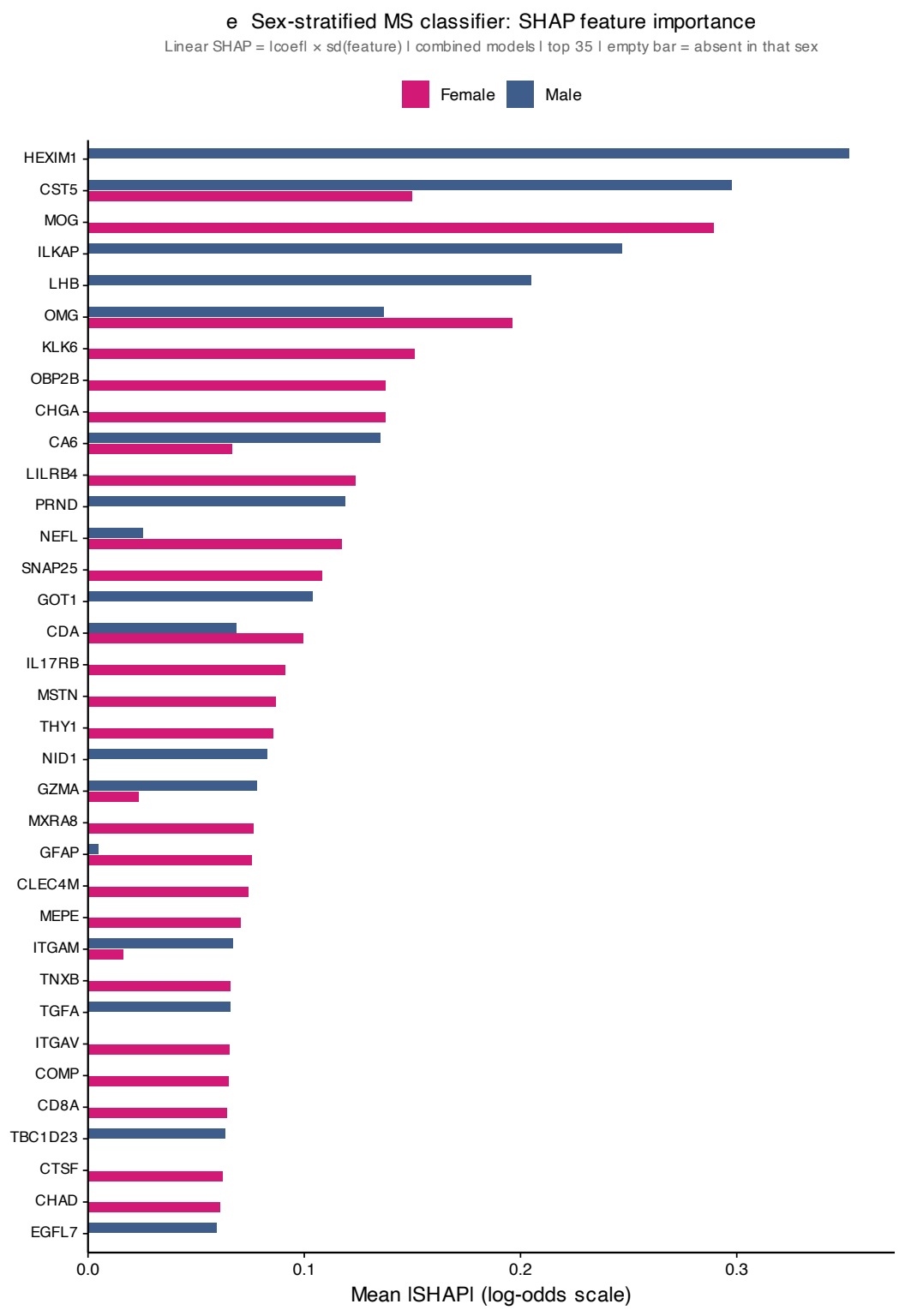
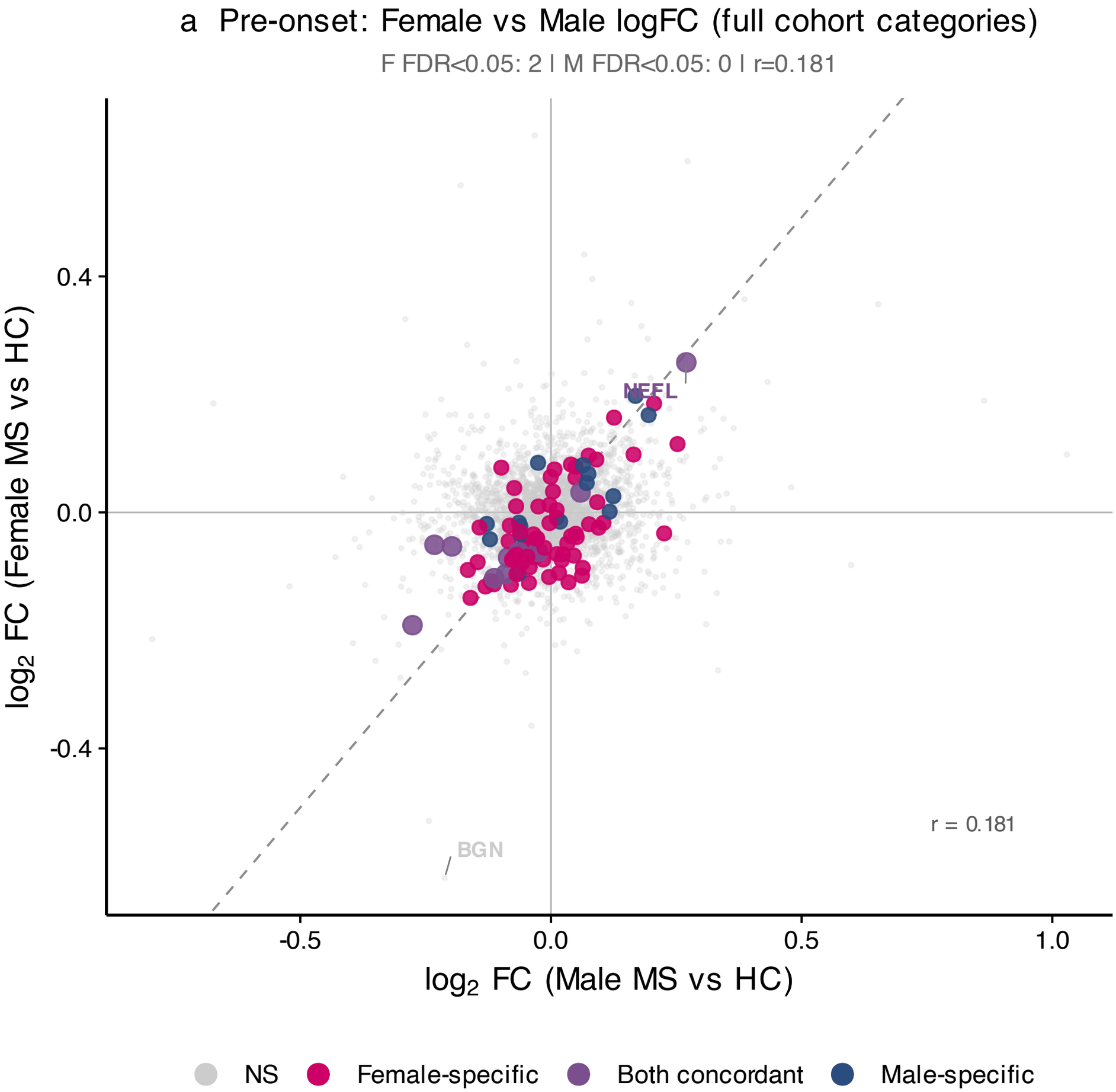

a Sex × MS interaction volcano
Interaction: sex:ms_status | 0 FDR<0.05 | 180 nominal p<0.05
SLC44A4
LHB
PSCA
OBP2B
TNNI3
DCTN2
CST7
FABP1
HRAS
VIM
ALPI
UGDH
0 FDR<0.05 | 180 nominal p<0.05
4
3
2
1
0
−log10 P
-0.6
-0.3
0.0
0.3
βsex×MS (interaction coefficient)

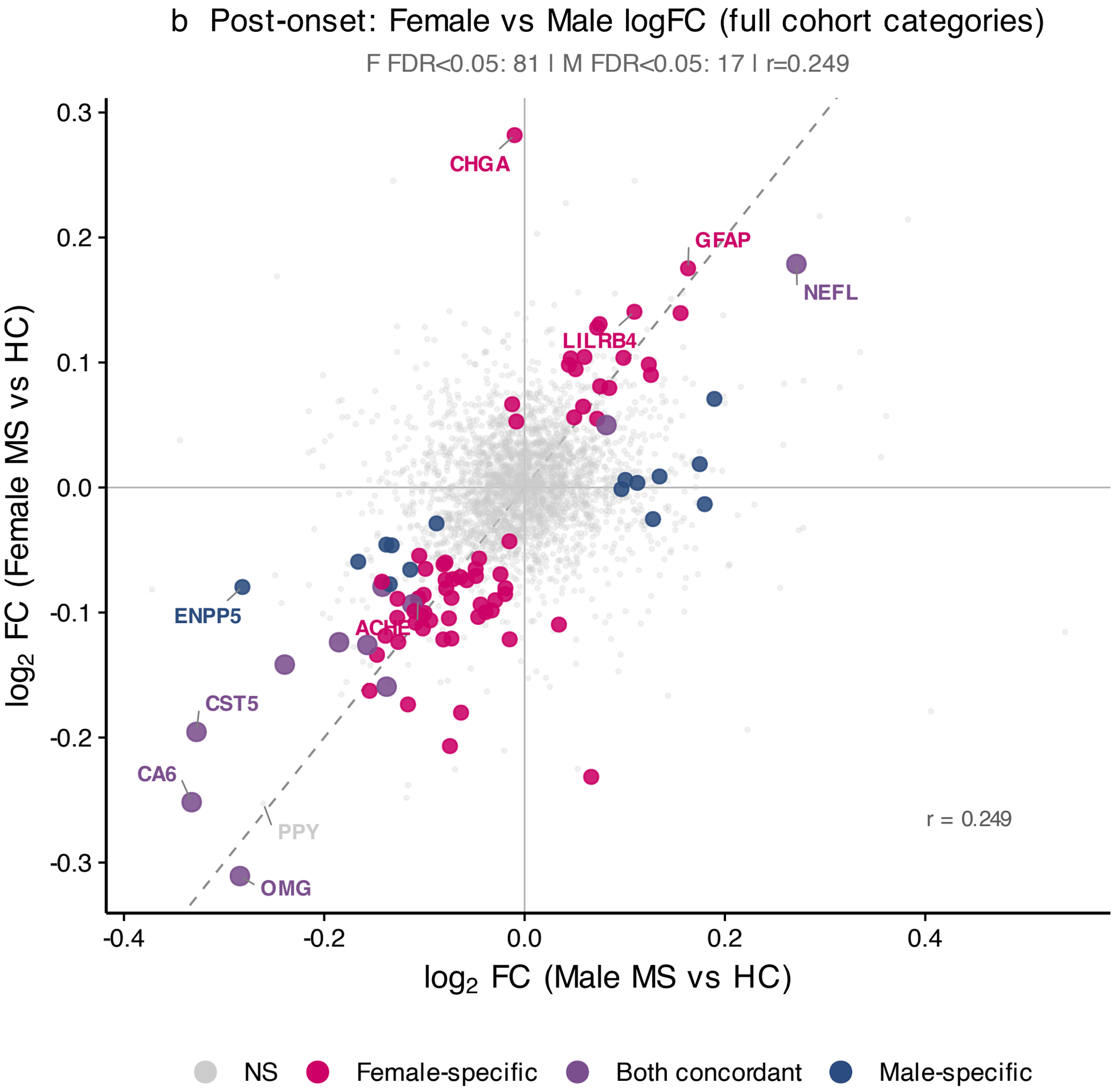

b Sex−stratified MS DEPs: Walchli+HPA cell−type expression
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Oligodendrocytes
Microglia
Astrocytes
Neurons
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
ITGB6
ITGA11
IGSF21
TGFA
IL17RB
COMP
BCAN
CA14
DNER
OMG
MSTN
NEFL
GFAP
OBP2B
LILRB4
RET
DPP4
CD8A
CHRDL2
ENPP5
FGFBP2
GZMA
CLEC4A
CST5
ITGAM
PEPD
BAG3
IL6
PLAUR
CDA
ARL13B
PI3
HPGDS
IGFBP2
CTSF
CHAD
GOT1
LAMP3
LPO
CA6
CLEC4M
CHGA
LRRN1
1
0.8
0.6
0.4
0.2
0
Category
Female−specific
Male−specific
Both concordant
Both discordant
NS
Compartment
Vascular
CNS
Immune


c Cell-type enrichment — F-MS vs HC
mma moderated-t (F-MS vs sex-matched HC) | NES>0: markers higher in F-MS | NES<0: lower | * FDR
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
F-MS
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4


d Cell-type enrichment — M-MS vs HC
mma moderated-t (M-MS vs sex-matched HC) | NES>0: markers higher in M-MS | NES<0: lower | * FDR
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
***
M-MS
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4


e Sex-stratified MS classifier: SHAP feature importance
Linear SHAP = |coef| × sd(feature) | combined models | top 35 | empty bar = absent in that sex
Female
Male
HEXIM1
CST5
MOG
ILKAP
LHB
OMG
KLK6
OBP2B
CHGA
CA6
LILRB4
PRND
NEFL
SNAP25
GOT1
CDA
IL17RB
MSTN
THY1
NID1
GZMA
MXRA8
GFAP
CLEC4M
MEPE
ITGAM
TNXB
TGFA
ITGAV
COMP
CD8A
TBC1D23
CTSF
CHAD
EGFL7
0.0
0.1
0.2
0.3
Mean |SHAP| (log-odds scale)


a Pre-onset: Female vs Male logFC (full cohort categories)
F FDR<0.05: 2 | M FDR<0.05: 0 | r=0.181
NEFL
BGN
r = 0.181
0.4
0.0
-0.4
-0.5
0.0
0.5
1.0
log2 FC (Female MS vs HC)
log2 FC (Male MS vs HC)
NS
Female-specific
Both concordant
Male-specific


b Post-onset: Female vs Male logFC (full cohort categories)
F FDR<0.05: 81 | M FDR<0.05: 17 | r=0.249
CHGA
GFAP
NEFL
LILRB4
ENPP5
ACHE
CST5
CA6
PPY
OMG
r = 0.249
0.3
0.2
0.1
0.0
-0.1
-0.2
-0.3
-0.4
-0.2
0.0
0.2
0.4
log2 FC (Female MS vs HC)
log2 FC (Male MS vs HC)
NS
Female-specific
Both concordant
Male-specific